Large language models for self-administered clinical vignette assessment: a pilot and validation study in Vietnam

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Short title: AI-based clinical vignette assessment in Vietnam

Abstract

We developed and validated a self-administered clinical vignette platform powered by a large language model (LLM), deployed through SurveyCTO, to measure primary health care provider competencies in Vietnam. In a pilot focus group, nine physicians rated LLM-simulated patient interactions as realistic (mean 3.78/5) and user-friendly. In the validation phase, 22 providers completed 132 vignette interactions across ten clinical scenarios. Essential diagnostic checklist scores – human-coded from transcripts – correlated with expert clinician evaluations (Pearson's $\rho = 0.49$–0.59). LLM-automated coding of checklist items from transcripts was feasible but noisy relative to human coding ($\rho = 0.35$). Multiple-choice knowledge test scores showed no significant correlation with vignette performance, suggesting the tool measures distinct dimensions of competency. The total cost of 132 chatbot interactions was under USD 2. LLM-driven conversational vignettes represent a low-cost and scalable method for assessing provider competencies, eliminating the need for enumerators while preserving the open-ended format critical to vignette validity. The platform is open-source and designed for replication in other health system contexts.

Author summary

Measuring the clinical skills of healthcare providers is essential for improving the quality of care, but current survey methods are expensive and require trained enumerators to travel to health facilities in person. We developed a new approach that uses large language models – the artificial intelligence technology behind tools like ChatGPT – to simulate patients in realistic clinical conversations that healthcare providers can complete on their own phones or laptops. In Vietnam, we tested this tool with 31 physicians across ten clinical scenarios. Providers found the simulated patient conversations realistic and easy to use. When we compared the results from our tool against expert physician ratings of the same conversations, the scores agreed well, suggesting the tool produces meaningful measures of clinical skill. We also tested whether AI could automatically score the conversations, which was feasible but noisy. This tool costs less than two US dollars for over a hundred consultations and requires no in-person surveyors, making it potentially transformative for routine, large-scale monitoring of healthcare quality in resource-limited settings. The platform and code are openly available for adaptation.

1 Introduction

Human resources for health represent the majority of total health expenditures in most countries, yet investments in primary health care education and training have not consistently translated into improved clinical competencies [1, 2]. Clinical vignettes – open-ended roleplay interactions with standardized patients – have emerged as a primary method for measuring provider competencies across health systems [3, 4, 5]. By eliciting each provider's unprompted approach to diagnosis and management of simulated cases, vignettes measure the best possible standard of care that providers are capable of delivering [6, 2]. However, traditional vignette surveys require two trained enumerators per interaction, significant travel budgets, and in-person coordination, making them expensive to deploy at scale or repeat at frequent intervals [5]. Digital multiple-choice formats undermine validity because unprompted elicitation of knowledge is the foundation of the method [3].

Recent advances in large language models (LLMs) have created new possibilities for scalable clinical assessment. LLM-based virtual patients can simulate realistic patient-clinician dialogues and provide automated performance feedback [7], and a rapidly growing literature has explored their use in medical education [8]. However, most existing applications target medical student training in high-income settings, and few have been validated against established competency measures or deployed for health workforce assessment in low- and middle-income countries, where the need is greatest [9]. Automated scoring of clinical interactions using LLMs has shown promise but remains noisy, with studies reporting moderate agreement with human evaluators at best [10, 11]. A tool that combines LLM-based patient simulation with automated data extraction – validated against expert clinical judgment – could enable routine, low-cost competency measurement without in-person enumerators.

This study presents such a tool: a new method for self-administered and automatically coded clinical vignettes using LLMs. We developed an LLM-based chatbot interface [12] using web-based SurveyCTO survey software compatible with mobile phones (**Figure 1**). In Vietnam, we piloted and validated this platform with primary health care providers across ten clinical scenarios. The chatbot – powered by OpenAI (gpt-4.1-nano) – role-played clinical cases in Vietnamese via respondents' personal mobile devices at a cost of under USD 2 for 132 interactions. We also developed an automated data extraction pipeline using Anthropic's Claude language model (claude-haiku-4-5) to classify provider performance from the resulting transcripts. We report results from (1) a focus group of 9 providers assessing usability and realism, and (2) a validation sample of 22 providers (132 interactions) comparing human-coded, LLM-coded, expert-graded, and knowledge-test-based competency measures.

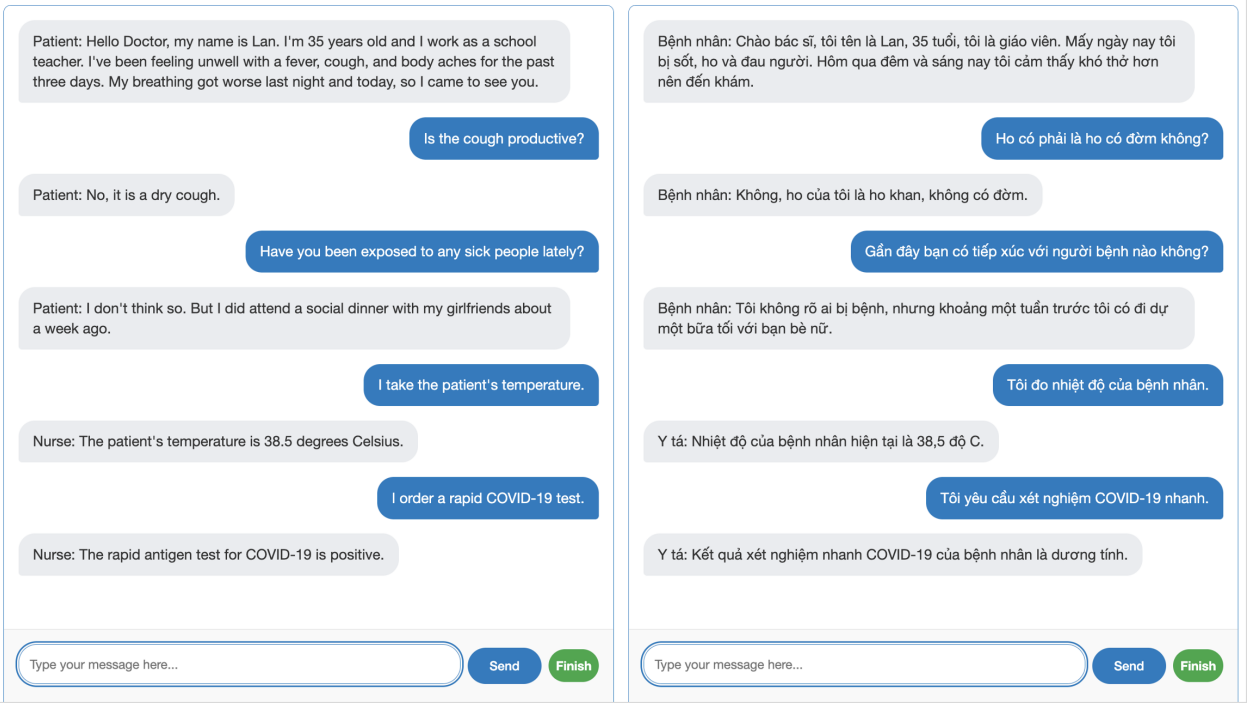

Fig 1: Screenshot of the LLM-based chatbot interface deployed through SurveyCTO. Left: English translation. Right: Original Vietnamese. The provider (blue bubbles) asks clinical questions and orders examinations; the LLM-simulated patient (grey bubbles) responds according to a pre-scripted clinical scenario. The interface runs on any smartphone or laptop browser.

2 Results

2.1 Focus group

Seven of the nine providers completed six cases within the allotted 45-minute focus group session, with average interaction time ranging from 3.4 to 9.2 minutes per case. All interactions were conducted in Vietnamese on smartphones. Across all conditions, respondents rated overall case realism positively, with an average of 3.78/5 (95% CI: 3.65–3.90).

Qualitative feedback highlighted several notable strengths. Providers consistently emphasized the ease of use and user-friendliness of the LLM-based conversational vignettes: *"Compared to other research tools, this one has a simpler interface and is easier to operate."* Fast and contextually appropriate responses were the most frequently cited benefits (4 mentions), with providers noting that information provided was sufficient and meaningful for clinical decision-making: *"The virtual patient's responses are relatively accurate, realistic, and easy to work with."* Others commended its realism, including language localization and the ability to return to history-taking after laboratory results were delivered: *"The language has been fully localized into Vietnamese and [is] very understandable"*; *"It doesn't give us everything at once for us to guess. The more*

we ask, [...] the more accurate the diagnosis will be." Another physician summarized: *"Regarding the system's responses [...], the level of realism is good. If we guide or structure our questions clearly, it will deliver as expected."*

A few respondents noted minor issues. The most common limitation was that certain laboratory or imaging tests were not performed or had results omitted (4 mentions). *"Some para-clinical tests weren't done, but when I asked a few related questions, [the responses] felt complete."* Although the limited test results raised concerns regarding realism, this was an intentional design feature to ensure reasonable standardization of the virtual patients: tests deemed irrelevant for the case were prompted to have results reported as "normal" or "unremarkable." The absence of non-verbal cues was also considered a barrier to realistic clinical interaction: *"When we see patients in person, we can quickly evaluate whether there is an infectious syndrome. For asthma patients, just by observing their breathing we can tell how severe their respiratory distress is. But in a simulation, we are limited in the questions we can ask."*

Respondents recommended several enhancements for future development. Suggestions included integrating visual elements such as laboratory images and video patient profiles, adding automatic performance feedback, and eventually transitioning from a web-based platform to an independent, offline smartphone application. One respondent also mentioned the potential benefit of including more complicated case scenarios to match clinical reality: *"In real hospitals, most patients don't just have one disease – they usually have multiple comorbidities."* Overall, the focus group participants judged the LLM-based vignette survey to be both usable and realistic.

2.2 Validation sample

Twenty-two providers completed 132 interactions in the validation sample (**Table 1**, **Table 2**). The average interaction lasted 10.19 minutes (95% CI: 9.03–11.35), excluding outliers lasting longer than one hour. The typical interaction resulted in the completion of 18.66% of the essential diagnostic checklist items for that case scenario (95% CI: 16.56–20.75). This low completion rate reflects the deliberately comprehensive nature of the checklists (30–55 case-specific items plus 19 common items per case; **Table 1**), which are designed to capture the full range of possible diagnostic actions rather than a minimum standard; similar checklist-based studies have reported comparable completion rates, as providers typically focus their history-taking on the most clinically relevant questions rather than exhaustively covering all possible items [3, 2]. Based on the Vietnamese transcripts, the first clinical expert rated the average interaction 3.42 on a 1–5 scale (95% CI: 3.24–3.59); based on the English transcripts, the other expert rated the average interaction 3.21 (95% CI: 3.05–3.37). The two expert ratings agreed well, with some variation (Pearson's $\rho = 0.62$, ordinary least

squares [OLS] $r^2 = 0.38$). Compared with the human-coded score on the essential diagnostic checklist, the expert ratings had Pearson's $\rho = 0.49$ and 0.59 for Vietnamese and English, respectively. In regression, a 10 percentage point (p.p.) increase in the checklist completion corresponded with an approximate 0.45-point increase in the expert rating. **Figure 2** illustrates these correlations.

Table 1: Clinical case scenarios used in the vignette survey. Each case includes case-specific checklist items plus 19 common items (clinical examination and medication questions administered across all cases).

Case	Condition	Category	Checklist items
1	Asthma	General	43
2	Pneumonia	General	30
3	Type 2 diabetes	General	39
4	Pulmonary tuberculosis	General	39
5	Hypertension	General	37
6	Hepatitis B (chronic)	Hepatitis	55
7	Hepatitis B (pregnancy)	Hepatitis	43
8	Hepatitis B (vaccination)	Hepatitis	39
9	Hepatitis C	Hepatitis	45
10	HIV/antiretroviral therapy	Hepatitis	44
	Common items (all cases)	–	19

Table 2: Validation sample summary statistics ($n = 132$ provider–case interactions from 22 providers completing six randomly assigned cases each).

Measure	Mean (95% CI)
Interaction duration (minutes)	10.19 (9.03–11.35)
Checklist completion (%)	18.66 (16.56–20.75)
Expert rating, Vietnamese (1–5)	3.42 (3.24–3.59)
Expert rating, English (1–5)	3.21 (3.05–3.37)

The LLM-automated coding was less strongly correlated with the human coding compared to the expert evaluations, and had substantially more statistical noise (Pearson's $\rho = 0.35$; OLS $r^2 = 0.12$). The correlation was statistically significant in regression against the human ground truth, with a 10 p.p. increase in the human-coded essential diagnostic checklist predicting a 1.7 p.p. increase in the LLM-coded checklist (**Figure 2**, Panel C). Finally, the multiple-choice question (MCQ) knowledge score for matched hepatitis cases had no statistically significant correlation with the essential diagnostic completion for the same cases (Pearson's $\rho = 0.07$; OLS $r^2 < 0.01$; **Figure 2**, Panel D).

The accuracy of the LLM coding of the essential diagnostic items varied widely. **Figure 3** plots the sensitivity and specificity of the LLM coding for 304 items for which both were calculable. The mean sensitivity was 0.31 (95% CI: 0.26–0.35); the mean specificity was 0.85 (95% CI: 0.83–0.87). There were three major clusters of items. First, many items had zero sensitivity, most of which had very few positives in the data; a large cluster of those had perfect specificity, indicating that the LLM accurately predicted "No"

for these items each time. Another cluster had perfect sensitivity and varied widely in specificity. Notably, among the remaining items, most were above the line of unity.

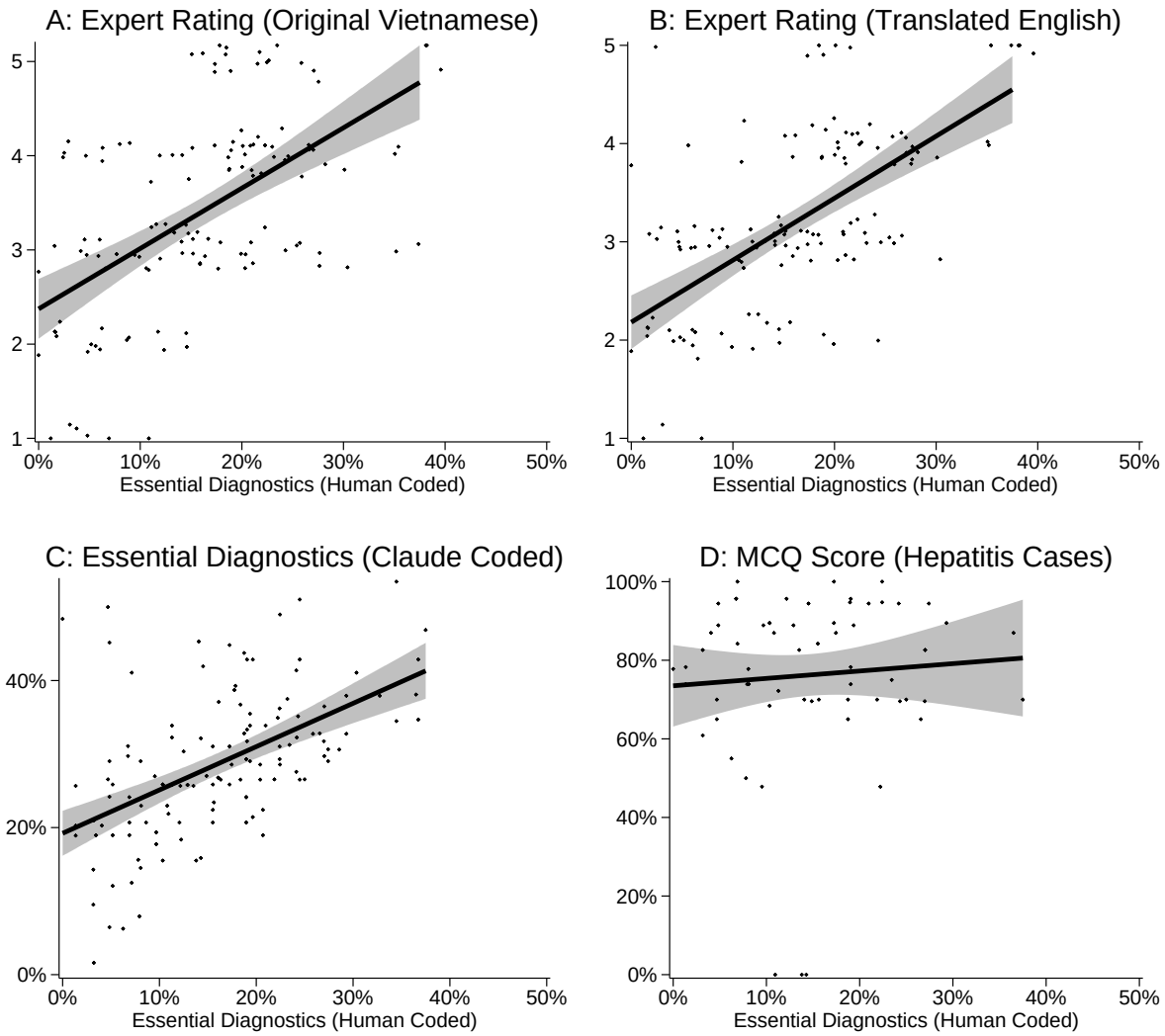

Fig 2: Validation of essential diagnostic checklist scores. Panel A: Expert rating in original Vietnamese vs. human-coded checklist completion. Panel B: Expert rating in translated English vs. human-coded checklist completion. Panel C: Claude-coded checklist vs. human-coded checklist. Panel D: Multiple-choice knowledge score vs. human-coded checklist (hepatitis cases only). Each point is one provider–case interaction. Lines show OLS fit with 95% confidence interval.

3 Discussion

Our results demonstrate that LLM-driven conversational vignettes produce competency measures that correlate well with expert evaluations of provider performance (Pearson's $\rho = 0.49$–0.59). The method eliminates

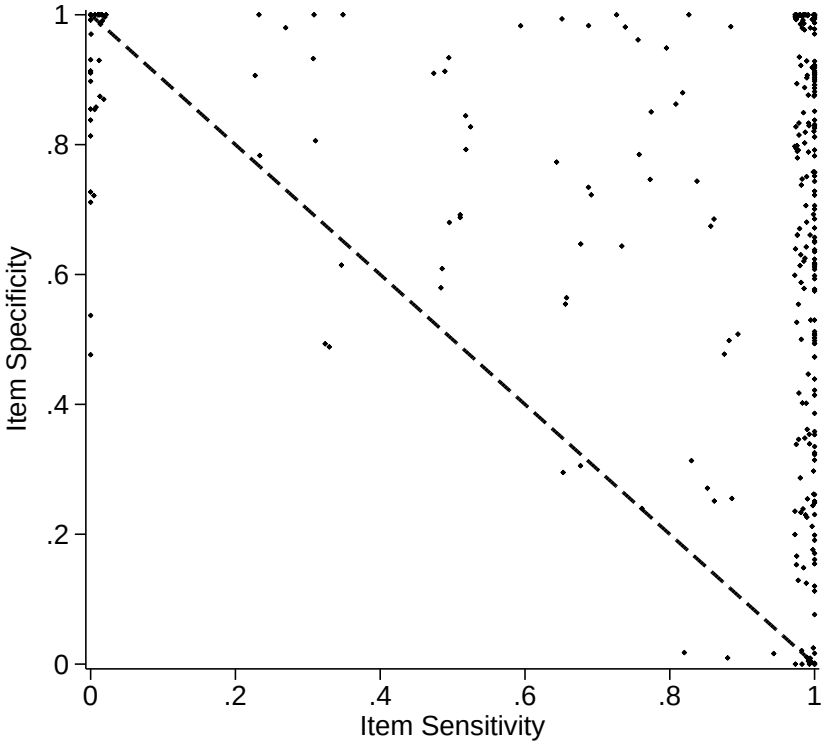

Fig 3: Item-level sensitivity and specificity of LLM coding against human coding. Each point represents one checklist item. Items above the dashed line of unity have specificity exceeding sensitivity. Items are included only where both metrics are calculable ($n = 304$).

the need for in-person enumerators while preserving the open-ended interaction format that is the foundation of vignette validity [3]. The platform's practical utility is substantial: 132 clinical interactions were completed at a total chatbot cost of under USD 2, with no in-person support beyond a brief Zoom orientation. Including human coding of transcripts, the total cost of the validation exercise was approximately USD 15 per provider – an order of magnitude less than traditional in-person vignette surveys, which typically require two trained enumerators per interaction plus travel and per diem costs [5].

We found that the Anthropic Claude LLM was able to extract structured data from transcripts to our specifications, and that these data correlated weakly but significantly with human-coded ground truth (Pearson's $\rho = 0.35$; OLS $r^2 = 0.12$). The low mean sensitivity (0.31) suggests that the LLM frequently fails to detect actions that providers actually performed, which would systematically underestimate competency if used without human review. This pattern is consistent with recent benchmarks of LLM-based clinical scoring, where automated systems achieve moderate agreement with human evaluators but struggle with nuanced interpretation of free-text responses [10, 11]. While LLM-assisted translation and human data extraction was not excessively burdensome at this scale, further research on the reliability and cost of automated data extraction may streamline data encoding at larger scales. Automated data extraction also preserves complete data in the form of transcripts and could enable future refinement or modification of checklists to fit the research purpose, rather than pre-encoding the data structure. Future research might also assess more expensive models, such as modern "reasoning" models, or develop a multi-shot "voting" algorithm to overcome some of its variability.

Our approach extends the emerging literature on LLM-based virtual patients [8, 7] in two important ways. First, whereas most existing applications target medical students in high-income educational settings, we demonstrate feasibility for health workforce assessment among practicing providers in a low- and middle-income country context. Second, we validate the resulting competency scores against independent expert clinician ratings and a traditional knowledge test, providing evidence that LLM-based vignettes measure meaningful dimensions of clinical competence. The null correlation between vignette performance and multiple-choice knowledge scores ($\rho = 0.07$) is consistent with earlier findings that open-ended vignettes capture diagnostic reasoning processes distinct from factual recall [3], though we note that this finding is based on a small sample (22 providers) and should be interpreted with caution. Future directions for tool refinement may include incorporating audio or video patient profiles to enhance realism through non-verbal cues.

The need for scalable competency measurement is acute. Evidence from sub-Saharan Africa [2], South and Southeast Asia [13, 14, 15, 16], and Vietnam specifically [17, 18] demonstrates that delivered care quality remains low despite growing investments in medical education, and that progress has been slow. The most

recent comprehensive assessment of primary care quality in Vietnam using clinical vignettes was conducted in 2015 [17]; the expense of replication has prevented a post-pandemic update. Tools like the one presented here could enable continuous, routine monitoring at a fraction of the cost of traditional approaches.

3.1 Limitations

This study has several limitations. First, the sample size was small (22 providers, 132 interactions), which limits the precision of our validation estimates and may not capture the full range of provider behavior. Second, the study was conducted in a single country (Vietnam) with a specific set of clinical conditions; generalizability to other settings, languages, and health systems requires further investigation. Third, the LLM-automated coding of checklist items was noisy relative to human coding ($\rho = 0.35$) and is not yet reliable for individual-level competency assessment. The low sensitivity (mean 0.31) suggests the LLM frequently fails to detect actions that providers did take, which would systematically underestimate competency. Fourth, we validated against human-coded checklists and expert ratings, but did not compare against direct clinical observation; the well-documented "know-do gap" means that vignette-based competency measures may not fully reflect actual practice [19]. Fifth, the text-based interaction format inherently lacks non-verbal clinical cues (e.g., visual inspection, auscultation sounds), which may constrain the types of clinical reasoning that can be assessed. Sixth, LLM behavior may vary across model versions and over time, which could affect reproducibility; we report specific model versions to facilitate replication. Finally, the platform requires internet connectivity and smartphone access, which may limit deployment in the most resource-constrained settings.

3.2 Conclusion

This study demonstrates that LLM-driven conversational vignettes can produce a low-cost, flexible, and scalable platform for measuring health care provider competencies. The method eliminates the need for in-person enumerators while preserving the open-ended interaction format that is the foundation of vignette validity. The platform is open-source, built on widely used survey software, and designed for adaptation to a wide variety of clinical contexts and languages. As LLM capabilities continue to improve, both the chatbot interactions and the automated data extraction are likely to become more reliable, further reducing the cost and increasing the feasibility of routine competency measurement. With further validation across settings and languages, this method has the potential to support rapid, low-cost measurement of health care provider competencies in diverse health system contexts.

4 Materials and methods

4.1 Study design and participants

In August 2025, we administered a web-based SurveyCTO survey (either by laptop or by mobile phone) that included LLM-based virtual patient "chatbots" preloaded with a pool of ten clinical cases (five general primary care conditions and five hepatitis-related) (**Table 1**). Each case scenario – detailing the patient's socioeconomic background, chief complaint, medical history, clinical exam results, and potential responses to a set of "checklist" questions – was scripted by expert clinicians supporting the research team. The OpenAI application programming interface (API; model gpt-4.1-nano) was prompted through the SurveyCTO software to perform according to these scenarios such that reasonable standardization was achieved across interactions. The scenarios developed for this study are available in the **Supporting Information**.

The research team first recruited and consented 9 physicians from a Provincial Hospital in Phu Tho, Vietnam, to a focus group for usability testing and UI/UX refinement. These providers received a live demonstration by the research team and were instructed to complete as many case scenarios as they found reasonable in a 45-minute period, followed by closed-ended (Likert scale) feedback questions on the realism of each case presentation. Two days later, the team conducted a structured group discussion to collect open-ended feedback, which was incorporated into the validation round.

For the validation sample, the research team contacted and consented 22 providers enrolled in the 2024 cohort of the StITCH hepatitis care training program, which co-designed and delivered a people-centered model of care for hepatitis B and C at the primary health care level in Vietnam. The validation group was given a Zoom orientation with no additional in-person support. These providers were assigned to complete six vignettes selected at random (three general cases and three hepatitis cases each). Afterwards, they completed a traditional multiple-choice knowledge test on hepatitis care, which included clinical knowledge questions based on the same five hepatitis scenarios. Although the total number of providers included in the validation sample was small, they completed 132 interactions containing 433 unique data points, allowing sufficient independent variation to assess the relationships between performance measures given that we do not engage in statistical hypothesis testing.

4.2 Chatbot implementation

The chatbot interface was built using the SurveyCTO platform's web-based survey tools with an LLM plugin [12]. Each clinical case was encoded as a structured system prompt (1,500–3,000 words) containing: the patient's demographic background and presenting complaint; a comprehensive medical and social history;

pre-scripted responses to anticipated history-taking questions organized by clinical domain; clinical examination findings to be revealed when specific examinations were ordered; laboratory and imaging results for relevant tests; and the correct diagnosis with recommended treatment and counseling points. To maintain standardization, the system prompt instructed the model to report results as "normal" or "unremarkable" for tests not specified in the case scenario, and to stay in character as the patient throughout the interaction. The full set of case prompts is available in the **Supporting Information**.

The OpenAI model gpt-4.1-nano was selected for cost-efficiency (approximately USD 0.015 per interaction) and adequate performance in Vietnamese-language roleplay. The SurveyCTO plugin manages conversational state by appending each user message and model response to the conversation history, which is passed in full with each subsequent API call. Providers accessed the survey on their personal smartphones or laptops via a URL link. The chatbot responded in Vietnamese (or the provider's preferred language) and maintained conversational context throughout each interaction. Providers could ask history-taking questions, order clinical examinations and laboratory tests, and provide a diagnosis and treatment plan, all through natural-language text input.

4.3 Data extraction and coding

Interaction transcripts were translated from Vietnamese to English using Anthropic's Claude language model (claude-haiku-4-5) and manually coded by the research team to determine whether each provider: (1) asked each of the pre-specified checklist history-taking questions, (2) ordered the recommended clinical examinations and laboratory tests, and (3) offered the correct diagnosis. The transcripts were also coded using the Anthropic Claude model (claude-haiku-4-5) with an identical schema as the manual coding, asking for each item, "In the following text, if the user was a doctor, did they specifically [ask/order/prescribe]...". Using the human-coded and LLM-coded checklist items separately, a "checklist score" was calculated for each vignette response as the proportion of items completed.

4.4 Outcomes

For the focus group, primary outcomes were: (1) qualitative feedback on the LLM-based vignette survey obtained through the focus group discussion, and (2) quantitative ratings of scenario presentation realism. For the validation sample, the primary outcome was validation of the human-coded "essential diagnostic checklist" LLM vignette score via interaction-level correlations with: (1) expert clinician grading of the transcripts both in English and in the original Vietnamese; and (2) the hepatitis multiple-choice knowledge test scores for matched cases. Our secondary outcome was the reliability of the LLM-coded "essential

diagnostic checklist," assessed by correlation with ground truth from human coding.

4.5 Statistical analysis

For the focus group, the transcript was thematically analyzed by the research team to identify common themes for tool refinement. Provider perceptions of vignette realism were captured through structured survey questions rating overall realism, clinical presentation realism, and social/economic presentation realism. We calculated means and 95% confidence intervals for each.

For the validation sample, agreement between manual and automated data extraction was assessed by calculating the proportion of agreement, true positives, false positives, true negatives, and false negatives based on the human coding as ground truth. The two expert clinicians who developed the vignettes also scored provider performance on each transcript from 1–5 separately for history-taking, diagnosis accuracy, and management completeness (one using the Vietnamese original and one using the English translation). The hepatitis knowledge test performance was scored as the total number of correct responses out of the administered items. Agreement of the human grading with each of these measures was assessed through unadjusted ordinary least squares (OLS) regression. Clustering was not necessary in any case since we calculated results separately by scenario – no provider repeated the same scenario twice. This study is reported following the Standards for Reporting of Diagnostic Accuracy Studies (STARD) guidelines where applicable to the LLM-versus-human coding comparison, and the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for the observational validation design. Completed checklists are provided in the **Supporting Information**.

4.6 Ethical approval

All procedures were approved by the ethical review board of the Hanoi University of Public Health (025-375/DD-YTCC) and Mass General Brigham Institutional Review Board (Protocol Number 2023P003011). All participants provided written informed consent prior to participation.

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Author contributions

Conceptualization: BD, DD. **Data curation:** BD, WSZ, HN. **Formal analysis:** BD. **Funding acquisition:** BD, DD. **Investigation:** BD, WSZ, HN. **Methodology:** BD, DD. **Project administration:** BD, HN, DD. **Resources:** DD. **Software:** BD. **Supervision:** BD, DD. **Validation:** BD, WSZ. **Visualization:** BD. **Writing – original draft:** BD. **Writing – review & editing:** BD, WSZ, HN, DD.

Data availability statement

The analytical code, structured rubric data, and de-identified aggregated data supporting the findings of this study are publicly available at `https://github.com/bbdaniels/vn-llm-pilot`. Individual-level transcript data cannot be shared publicly because they contain identifiable clinical interaction content from a small number of participants who consented to research use only; access to these data may be requested through the Mass General Brigham Institutional Review Board (Protocol 2023P003011).

Competing interests

The authors declare no competing interests.

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Supporting information

S1 Appendix. Clinical case scenarios and essential diagnostic checklists for all ten vignette cases, supplementary tables, and focus group supplementary results.

Table S1. Case scenarios and realism ratings from the focus group, including opening statements, recommended treatment, and mean realism ratings across four dimensions.

Table S2. Summary of provider performance by case in the validation sample, including checklist completion, correct diagnosis rates, and interaction duration.

Figure S1. Average realism rating by condition and feedback question in the focus group.

Figure S2. Average case duration by condition in the focus group.

Figure S3. Distribution of focus group realism ratings by feedback question.

S1 Checklist. STROBE checklist for observational studies.

S2 Checklist. STARD checklist for diagnostic accuracy studies.